

PDE C1 Finite-Galerkin Separation — Self-Contained Review Bundle

This is one file containing the whole reviewer packet, assembled by `scripts/build-c1-review-bundle.mjs` from the on-disk artifacts. It is the attachment the outreach email refers to as the packet. Send this (or its PDF rendering) plus the email from `docs/proof/PDE_C1_EXTERNAL_REVIEW_EMAIL_DRAFT.md`.

Status: PROVISIONAL / UNPROMOTED, gated on external review. This is a structural separation in a finite-dimensional Galerkin truncation of 2D Navier–Stokes. It is **not** a Navier–Stokes existence/smoothness claim, **not** a new determining-modes theorem, **not** a statement about the infinite-dimensional NSE attractor, and **not** a Clay-problem claim.

The one-sentence ask

On the sampled invariant measure of a 2D Kolmogorov-flow Galerkin model ($k_f = 2$, $K = 3$, $G \in \{200, 300\}$), the low-band signature Φ_K is **state-insufficient** (certified non-injective) yet **control-sufficient** for a registered decision (Φ_K -measurable up to a measure- $\delta \approx 0.037$ boundary layer). Is this a genuine finite-Galerkin separation between state-reconstruction and action-sufficiency, or ordinary observer / data-assimilation / LES-closure behavior that should not be made load-bearing?

The four locked questions

- State-insufficiency language.** Given the twin-state certificate is sampled-SRB / finite-Galerkin (not an exact attractor theorem), is " Φ_K non-injective / state-insufficient on $\text{supp } \mu$ " reasonable, or should it be weakened to a finite-sample claim?
- Control-sufficiency language.** Is the kNN/disintegration reading of " π^* factors through Φ_K up to μ -measure zero" (local action-mixing $\text{mean_minority} \rightarrow 0$, plus paired action-constancy on the certified non-injective pairs) mathematically fair, or overstated?
- Objective legitimacy.** Is a held-out look-ahead-max quantile a defensible regime-portable proxy action, given the older burn-in-percentile trigger went vacuous at $G = 300$ and was replaced (not retuned)?
- Real separation vs renaming.** A genuine "a sub-determining mode set can be control-sufficient without being state-reconstructive" separation, or ordinary functional-observability / data-assimilation / LES-closure behavior?

The answer menu (a paragraph is plenty — a negative reply is the most valuable)

The framing seems conservative / basically right.
Weaken or rename X (e.g. "state-insufficient" $\rightarrow$ finite-sample only).
The control-sufficiency / fiber language is too strong because Z.
This is standard observer / data-assimilation, cite B.
The objective is implicitly a function of Φ_K ; the separation is vacuous because W.

Anti-folklore guard (worth pre-empting)

The natural reviewer reflex — "this is just LES / closure: of course coarse energy is predictable from coarse state" — is exactly what the non-injectivity certificate is meant to foreclose: LES / AIM / closure assume or derive **state-sufficiency** via slaving; C1 *certifies state-insufficiency* (twin states) and shows the **decision** survives the genuinely unresolved fine state. Please stress-test that distinction specifically.

Read order below: separation statement → result → two adjudicators → three run receipts (verbatim). Each section is the unmodified on-disk artifact.

[PRIMARY · separation statement]

source: docs/proof/PDE_C1_SEPARATION_STATEMENT.md · sha256:98eea9515f52

PDE C1 — Finite-Galerkin Structural Separation (reviewer-facing statement)

Proof-track consolidation, 2026-05-29. Lifts the Postulate-1 reading (PDE_DETERMINING_MODES_POSTULATE1.md) into a single self-contained **finite-Galerkin** separation statement with its empirical witness, an anti-vacuity ledger, and a determining-modes comparator. This is the artifact to put in front of a PDE reviewer.

Scope. This is a structural separation in a finite-dimensional dynamical system that is a Galerkin truncation of 2D Navier-Stokes. It is **not** a Navier-Stokes existence/smoothness claim, **not** a new determining-modes theorem, and **not** a statement about the infinite-dimensional NSE attractor. Status: **PROVISIONAL, UNPROMOTED**, gated on external review.

1. The statement

Let S_N be the 2D Kolmogorov flow truncated to a fixed Galerkin cutoff ($N = 16$, 32×32 grid, pseudo-spectral vorticity), forced at $k_f = 2$, at Grashof number G . Let μ be the empirical (sampled SRB-like) invariant measure on its attractor. Let

```
Phi_K : state -> R^d      d = 18  (the K = 3 low-Fourier band, 9 complex modes)
```

be the low-band signature, and let $\mathcal{J}_q$ be the registered control objective (Section 3). Write π^* for the $\mathcal{J}_q$ - optimal binary selector ({no_op, damp_low_band}).

Claim (regime-2 separation). On (S_N, μ) at $G \in \{200, 300\}$, $k_f = 2$, $K = 3$:

1. **State-insufficient.** Φ_K is **not injective** on $\text{supp } \mu$: there is positive μ -mass of state pairs with Φ_K -distance $\leq \epsilon_K$ and complementary high-mode (Q_K) separation $\geq \delta_H$. (*twin-state certificate, CERTIFIED both regimes*)
2. **Control-sufficient.** π^* factors through Φ_K up to μ -measure zero: on ϵ_K signature-balls the proxy action is near-constant ($\text{mean}_{\text{minority}} \rightarrow 0$), and on the **certified non-injective pairs themselves** the action disagreement $D_{\text{witness}} \leq \delta_{\text{action}}$. (*kNN convergence POSITIVE + paired fiber-constancy exhibit*)

The smallest control-sufficient signature for J_q is therefore strictly smaller than any state-reconstructive signature on this support: Φ_K collapses states that determination would still separate, yet those collapsed states share the J_q -optimal action.

This is precisely Reading-2 regime 2 of the Postulate-1 sidecar — "state insufficient, control sufficient" — the non-vacuous target — now stated as a finite-dimensional fact with a pre-registered numerical witness.

2. Decision domain, signature, measure

Object	This cell
Decision domain X	Galerkin state of S_N ($d_{\text{state}} = 440$ real DOF)
Signature Φ_K	low-Fourier band $K = 3$, $d = 18$ real components
Null coordinate Q_K	complementary high modes, $d_{\text{high}} = 422$
Measure μ	empirical invariant measure, 50000 samples, interval 50 steps after 10^5 burn-in
Objective J_q	held-out quantile-calibrated low-band overshoot trigger (Section 3)
Selector π^*	$1[\text{lookahead}_{\text{max}}(\text{state}) > E_{\text{max}}]$, $E_{\text{max}} = \text{held-out } (1 - q)\text{-quantile}$

The signature resolves **18 of 440** real degrees of freedom (~4%); **422** are unresolved. That gross resolution gap is what makes non-injectivity plausible and the separation non-trivial.

3. The objective is not a function of the signature (anti-tautology)

The decisive anti-vacuity property: J_q is defined on the **full state's future**, not on Φ_K . The label is

$$a(x) = 1[\max_{t \leq \tau} E_{\text{low}}(\text{flow}_t(x)) > E_{\text{max}}],$$

a τ -step look-ahead of the *full nonlinear flow* flow_t , thresholded at a quantile E_{max} calibrated on a **disjoint** held-out window. So control-sufficiency — that this full-state look-ahead label is nonetheless Φ_K -measurable a.e. — is a **dynamical fact about s_N** , not a definitional consequence of how the objective was written. If the label were a function of Φ_K directly (e.g. instantaneous $E_{\text{low}} = ||\Phi_K||$ -band energy), the claim would be vacuous; it is not, because the label is a forward-time excursion of the full state.

4. Anti-vacuity ledger

Each way the separation could be trivial, and why it is excluded here.

Vacuity mode (from the sidocar)	Excluded because
Objective is full-state tracking / reconstruction	$\mathbb{J}_q$ is a scalar binary trigger, not state tracking.
Action directly actuates all unresolved components	Actions are <code>{no_op, damp_low_band}</code> ; neither reads/penalizes Q_K .
Selector separates every state-distinct fiber	Falsified by the paired exhibit: certified Q_K -separated pairs share the action ($D_{\text{witness}} \leq \text{delta_action}$).
Determining machinery already reconstructs at $\leq K$ modes	Falsified internally: Φ_K ($K=3$) is certified non-injective on this support (Section 5).
Strictness only appears after a post-hoc objective change	E_{max} is held-out quantile-calibrated and frozen before scoring; no post-hoc retune (would be PDE-C1-NEG-B).
Objective is constant on the whole support (degenerate)	$\text{damp_fraction} = 0.298$ ($G=200$) / 0.269 ($G=300$) — non-trivial; the harness delta_proxy_min / structural-constancy gates guard this.

The harness encodes the last three as hard gates (`DEFERRED_VACUITY` , structural-vacuity precedence, `PDE-C1-NEG-B` on post-hoc retune), so a vacuous instance defers or files negative rather than passing.

5. Determining-modes comparator (internal, no borrowed constant)

The state-reconstruction reference is supplied **internally** by a fixed- Galerkin bracket, not by importing an asymptotic literature constant (which would be loose at moderate G):

- **Lower bracket (have it).** At $K = 3$ the twin-state certificate is POSITIVE: signature-near pairs with Q_K separation $p_{50} = 0.0154$ at high-mode-norm median 0.233 (~6.6% of the norm) within ϵ_{K} -balls of radius ~ 0.06 . So $K = 3$ provably does **not** determine state on $\text{supp } \mu$ — a sampled-support determining-modes *negative*.
- **Upper bracket (registered next run).** Increase K until the twin-state witness vanishes (injectivity returns). The smallest such K^* is an internal, geometry-specific upper bound on the determining count for this cell; the separation lives in the gap $K = 3 < K^*$. Pre-registered as the K -window companion (also serves the robustness sub-goal).

Literature backdrop (qualitative). Foias-Prodi / Foias-Temam and Constantin-Foias-Manley-Temam establish that finitely many modes determine 2D NSE long-time dynamics, with the determining count finite and growing with G . We use this only as the reason a finite K^* exists; the *value* here is measured, not asserted.

6. Theorem vs. witness — what is and is not proved

- **Provable-as-witnessed (finite-dimensional).** On (S_N, μ) the three numerical predicates — non-injectivity, neighbourhood action-constancy, paired action-constancy — are pre-registered, gated, and reproduced deterministically (seed `20260528`). Within the sampled support they *certify* the regime-2 separation for this cell. This is an existence witness in a finite-dimensional dynamical system, not a theorem schema.

- **Not claimed.** (i) The infinite-dimensional NSE attractor — S_N is a truncation; the support is sampled, not the full attractor. (ii) Any Millennium-problem content. (iii) A new determining-modes bound. (iv) Generality beyond the tested objective family / forcing geometry ($k_f = 2$) — that is the robustness sub-goal, in progress.

7. Observer-theory restatement and prior-art positioning

The separation restates cleanly in control/observer language, which is where its novelty sits (recon: `PDE_C1_MECHANISM_RECON.md`). Take the Galerkin flow as the system, Φ_K as the measurement/output map, μ as the invariant measure. Then C1 says:

Φ_K is **decision-observable but state-unobservable** on $\text{supp } \mu$: the full state is *not* functionally observable from Φ_K (certified non-injectivity — the output map has positive- μ -mass fibers), yet the registered safety decision $a(x) = 1[\max_{t \leq \tau} E_{\text{low}}(\text{flow}_t x) > E_{\text{max}}]$ is Φ_K -measurable up to a measure- δ boundary layer ($\delta \sim D_{\text{witness}} \sim 0.037$).

Pivotal claim a reviewer must bless or reject (everything else is support): *on the sampled invariant measure of this Galerkin 2D-NSE system, the low-band projection Φ_K is non-injective (state-unobservable) yet the registered decision is Φ_K -measurable up to a measure- δ boundary layer (decision-observable), with K below the determining-modes count.*

Prior-art δ — why this is an extension, not a renaming:

Established	C1's δ
Functional observability is linear / finite-dim / network / estimation (Montanari–Motter PNAS 2022); the nonlinear extension (arXiv 2301.04108) lists turbulent observables like energy on spatially-extended systems as an open gap	Instantiated on a (Galerkin) NSE attractor , for energy (the named-open observable), for a decision objective (not estimation), with a measured non-injectivity certificate
Determining functionals determine the state (Cockburn–Jones–Titi 1997)	Φ_K is below the determining threshold (non-determining) yet decision-sufficient — the complement of that theory
Approximate inertial manifolds slave high→low, i.e. state- sufficient	C1 is below the AIM threshold — state- insufficient , certified — the opposite regime
Mori–Zwanzig closure : the unresolved coupling into the resolved energy budget is <i>approximable</i> via memory (Parish–Duraismy 2017)	C1 measures that the coupling R is ~99% signature-determined ($R^2(R \Phi_K) = 0.998$ ($G=200$) / 0.990 ($G=300$), at the exact-function ceiling, neg-control ~0) — a local energy-budget closure $R \approx R(\Phi_K)$ that holds <i>precisely where state reconstruction fails</i> ; this is the measured mechanism, not a fitted closure (see <code>PDE_C1_MZ_ENERGY_BUDGET.md</code> §10)

Anti-folklore guard. The natural reviewer reflex — "this is just LES / closure: of course coarse energy is predictable from coarse state" — is exactly what the non-injectivity certificate forecloses. LES / AIM / closure assume or derive **state-sufficiency** via slaving; C1 *certifies state-insufficiency* (twin states) and shows the **decision** survives the unreconstructable state anyway. That is the distinction between "coarse predicts coarse because coarse slaves fine" and "the coarse decision is robust to genuinely unresolved fine state."

8. Review path and cross-references

The three reviewer questions (sidecar Section "External Review Path") map to: (1) Reading 1 faithful to determining modes — Section 5 + lit-pass; (2) Reading 2 a real distinction — Sections 1, 3 (anti-tautology); (3) no hidden reconstruction in `Phi_K` — Section 4 ledger + Section 5 internal non-injectivity. The reviewer email is staged at `PDE_C1_EXTERNAL_REVIEW_EMAIL_DRAFT.md`.

- `PDE_DETERMINING_MODES_POSTULATE1.md` — the reading note this consolidates.
- `PDE_C1_TWIN_STATE_CERTIFICATE.md` — state-insufficiency half.
- `PDE_C1_KNN_CONVERGENCE_CHECK.md` — control-sufficiency half.
- `PDE_C1_PAISED_FIBER_CONSTANCY.md` — the paired exhibit composing them on the same pairs.
- `PDE_C1_REGIME_GENERALITY_v1.md` — the two-regime portable objective.
- `../SUNDOG_V_NAVIERSTOKES.md` — the ledger.

=====

[RESULT · two-regime portable objective]

source: `docs/proof/PDE_C1_REGIME_GENERALITY_v1.md` · **sha256:23f17da8bob6**

PDE C1 Regime Generality v1 — Portable Objective

Pre-registration and result, filed 2026-05-29. Successor to `PDE_C1_REGIME_GENERALITY_v0.md`, whose $G=300$ probe returned `PDE-C1-RG-DEFERRED_VACUITY`: the kNN/twin-state machinery stayed usable, but the *registered overshoot proxy went near-vacuous* (`damp_fraction` ≈ 0.004) under heavy-tailed/intermittent energy statistics. The regime-generality question was therefore **not adjudicated** — the failure was objective construction, not fiber-locality coverage or support-level non-injectivity. This artifact replaces the fixed-percentile objective with a **regime-portable** one and re-poses the Grashof-axis generality test cleanly.

Status: executed. §12 records `PDE-C1-RG-POS`: the portable objective passes its portability gate at $G=200$ and $G=300$, the control half is POSITIVE at both regimes, and the $G=300$ twin-state companion is CERTIFIED. C1 remains unpromoted because proxy faithfulness and external PDE review remain open.

1. The fix in one line

Replace "exceed the **burn-in** 95th-percentile `E_K` in lookahead" with "be in the top $(1-q)$ fraction of **own-regime** τ -excursions," calibrated on a **held-out post-burn-in** window. This pins `damp_fraction` $= 1-q$ by construction at every regime, removing the vo vacuity failure mode.

2. Why the vo objective was not portable

`E_max` was the 95th percentile of the (last-25%) burn-in `E_K`. Two coupled defects under intermittency:

- **Calibration/sampling mismatch.** A single burst in the burn-in calibration window shifts the 95th percentile far out; the post-burn-in sampling window is comparatively quiescent $\rightarrow$ almost no look-ahead window reaches `E_max`.

- **Heavy-tailed shape-dependence.** At $G=200$ the energy distribution is tight, so the 95th percentile sits near the bulk (damp_fraction 0.30); at $G=300$ it is intermittent, so the 95th percentile sits on a rare-burst tail (damp_fraction 0.004). The *same* percentile rule means different things at different regimes.

Confirmed regime property, not artifact: v_2 ($G=300$, full-burnin $E_{\max}$) also gave sub-1% damp. The objective, not the machinery, fails to transfer.

3. Portable objective (pinned)

For a state u with no-op τ -lookahead trajectory, define the **lookahead-max excursion**

$$M(u) = \max_{t \in [0, \tau]} E_K(u(t)), \quad \tau = 5.0 \text{ time units.}$$

Calibrate the threshold on a held-out post-burn-in calibration window C (disjoint from the adjudication sample A , see §4):

$$E_{\max} = \text{quantile}_q(\{M(u) : u \in C\}), \quad q = 0.70.$$

The proxy selector, applied to the adjudication sample A :

$$\begin{aligned} \text{pi_hat}(u) &= \text{damp_low_band} && \text{iff } M(u) > E_{\max} \\ &= \text{no_op} && \text{otherwise, for } u \in A. \end{aligned}$$

By construction damp_fraction on C is exactly $1 - q = 0.30$; on the disjoint stationary A it is ≈ 0.30 (the **portability gate**, §6, verifies this at both regimes). $q = 0.70$ is chosen so the new objective's damp scale matches v_5 's emergent 0.30, making the $G=200$ re-run a clean comparison to the existing witness.

Everything else is inherited from v_5/v_6 unchanged: $k_f = 2$, $K = 3$, $d_K = 18$, $dt = 0.01$, $\tau = 500$ steps, $\epsilon_K = 0.05 \cdot \sqrt{2 E_{\max}}$ (now with the portable $E_{\max}$), kNN sweep $k \in \{10, 15, 20, 25, 30, 40, 50\}$ with a_{mm} thresholds (≤ 0.005 POSITIVE, ≥ 0.015 NEG-A), twin-state $k_{\text{twin}}=50$, $\delta_H = \max(1e-6, 0.05 \cdot \text{median}\|Q_K\|)$, gates 0.01 / 100.

4. Held-out split (pinned)

The post-burn-in trajectory is partitioned into two disjoint blocks with a decorrelation gap:

```
calibration window C : 50,000 samples at interval 50 steps
decorrelation gap    : 5,000 steps (~a few Lyapunov times)
adjudication sample A: 50,000 samples at interval 50 steps
```

$E_{\max}$ is computed from C only; labelling, kNN convergence, and twin-state run on A only. A is held at 50,000 to match v_5 's adjudication N for apples-to-apples comparison. Total post-burn-in integration roughly doubles vs v_5 (~40 min/run expected; see §7 cost). Rationale for held-out rather than in-sample: although $E_{\max}$ is a single global scalar (so in-sample calibration would not obviously leak *fiber* structure into the local kNN test), held-out removes all doubt at low cost and matches the agreed design.

Cost-reduced alternative (not pinned; a sign-off option): $C = A = 30,000$ with the same gap, $\sim 1.2 \times v_5$ cost. Decide at sign-off.

5. Harness objective-mode

A new objective mode, selected by a flag, leaves the v0–v6 fixed- percentile path untouched:

```
--objective {overshoot-burnin (default), portable-quantile}
```

portable-quantile config additions (pinned in the preset, pre- registered): `objective_quantile q = 0.70`, `calibration_sample_count = 50000`, `calibration_gap_steps = 5000`. New presets `lock_v7_g200` (`k_f=2`, `G=200`, `K=3`) and `lock_v7_g300` (`k_f=2`, `G=300`, `K=3`), both with `e_max_burnin_fraction` irrelevant under this objective (`E_max` no longer from burn-in) and `objective = portable-quantile`.

Implemented by extending `run_cell` to integrate the extra calibration block + gap before the adjudication block; compute $M(u)$ for calibration samples; set `E_max = quantile(M_C, q)`; then label/adjudicate the adjudication block exactly as today. All adjudicators (`knn`, `knn-sweep`, `twin-state`) consume the adjudication block unchanged. Smoke parity and overshoot-burnin regression were checked before the lock runs.

6. Portability gate (new, pre-registered)

Before interpreting any control-sufficiency verdict, confirm the objective is actually portable:

```
damp_fraction(A) in [0.20, 0.40] at BOTH G=200 and G=300.
```

- If both pass → the objective is portable; proceed to read the kNN verdicts as a genuine regime comparison.
- If either fails → **PDE-C1-RG-PORTABILITY-FAIL**: even the held-out-quantile objective does not transfer (e.g. severe non-stationarity between `C` and `A`). This is itself a finding; do **not** retune `q` or the split to rescue it (that would be **PDE-C1-RG-NEG-B**). A new construction (burst-onset, §9) would be a separate v2 pre-registration.

`[0.20, 0.40]` is 0.30 ± 0.10 ; by construction `A` should sit very near `0.30`, so this gate mainly catches stationarity/leakage failures.

7. Program and run order (pinned)

```
1. lock_v7_g200 --adjudicator knn-sweep (positive control + de-confound)
2. lock_v7_g300 --adjudicator knn-sweep (the generality test)
   [portability gate checked on both before interpreting either]
3. IF both return STRICTNESS_WITNESS_POSITIVE:
   lock_v7_g300 --adjudicator twin-state (support companion)
   (lock_v7_g200 twin-state optional, for a complete re-witness)
```

The **G=200 re-run is mandatory and runs first**: the portable objective is a *new* objective, so the v5 witness must be re-established under it before G=300 generality means anything. If G=200 flips (does **not** return POSITIVE under the portable objective), that is a critical finding — it would mean the v5 POSITIVE was partly an artifact of the old fixed-percentile threshold — and the generality program pauses to reconcile before G=300 is interpreted.

Expected wall-clock ~40 min per kNN-sweep run ($\approx 2 \times$ v5 from the doubled post-burn-in integration), ~35 min per twin-state run. Up to ~4 runs. None inline under the ~10-minute rule.

8. Branches and interpretation

Per-regime kNN verdict feeds the existing `PDE_C1_REGIME_GENERALITY_v0.md` §6 branch family, gated by §6 portability here:

G=200 kNN	G=300 kNN	program outcome
POSITIVE	POSITIVE	PDE-C1-RG-POS (control-suff replicates across G under a portable objective); proceed to twin-state
POSITIVE	NEG-A	PDE-C1-RG-NEG-A (regime-2 is a localized window on the Grashof axis — the informative failure)
POSITIVE	deferral	PDE-C1-RG-INCONCLUSIVE_CONTROL
not POSITIVE	—	program pauses: portable objective overturns the v5 witness; reconcile before any generality claim

PDE-C1-RG-POS here means: the v5 Reading-2 regime-2 witness has a higher-Grashof replication **under an objective that is portable by construction** — strictly stronger than a same-objective coincidence. It still does **not** mean generality across all `G`, across `k_f`, an infinite-dimensional NSE theorem, `J`-optimality, or promotion without external review.

9. Pre-registration discipline

- All values in §3–§6 are fixed before any verdict-bearing run.
- Post-hoc change to `q`, the split, the gap, `epsilon_k`, the kNN/ twin-state thresholds, or the regime after reading a receipt → PDE-C1-RG-NEG-B.
- The portability gate is a **precondition**, not a tunable: a `PORTABILITY-FAIL` is filed, not rescued.
- Burst-onset (a `dE_k/dt` or relative-jump onset criterion) is the **documented v2 alternative**, not part of this artifact; it carries more free parameters and would need its own pre-registration.

10. Build Decisions Closed

The sign-off choices landed as:

1. **Split sizes:** 50k calibration / 50k adjudication, preserving apples-to-apples comparison with v5.
2. **Quantile:** `q = 0.70`, targeting `damp_fraction ≈ 0.30` to match the v5 action scale.
3. **Build:** `--objective portable-quantile` plus `lock_v7_g200` / `lock_v7_g300` presets, with regression and smoke checks before the verdict-bearing runs.

12. Result (2026-05-29) — PDE-C1-RG-POS

The full program executed; all three verdict-bearing runs landed. The portable objective fixed the v6 vacuity and the complete regime-2 witness **replicates at G=300**.

run	portability gate (adj damp $\in [0.20, 0.40]$)	verdict
lock_v7_g200 kNN (positive control)	0.3003 ✓ (calib 0.300)	STRICTNESS_WITNESS_POSITIVE , $a_{mm} = -0.00079$, slope 0.736
lock_v7_g300 kNN (generality test)	0.2688 ✓ (calib 0.300)	STRICTNESS_WITNESS_POSITIVE , $a_{mm} = +0.00058$, slope 0.564
lock_v7_g300 twin-state	—	TWIN_STATE_CERTIFIED (100% witness coverage, 942,834 unique pairs, $\delta_H = 0.0111$ from real median $\ Q_K\ = 0.222$)

Receipts: results/proof/c1-rg-v1-g200-knn-sweep/ , results/proof/c1-rg-v1-g300-knn-sweep/ , results/proof/c1-rg-v1-g300-twin-state/ .

What it establishes

- **Portability gate passed at both regimes.** Held-out $damp_fraction = 0.300$ ($G=200$) and 0.269 ($G=300$), both in $[0.20, 0.40]$. The portable objective is genuinely regime-portable: v6's $damp_fraction = 0.004$ vacuity at $G=300$ is gone (0.269 now). The small $G=300$ calib-vs-adj gap ($0.300 \rightarrow 0.269$) is a mild, honest signature of intermittency non-stationarity, well inside the gate.
- **Positive control ($G=200$) re-establishes the v5 witness under a different, sounder objective.** $a_{mm} = -0.00079$, slope 0.736 vs v5's $-0.00078 / 0.737$ — near-identical despite a different E_{max} construction (held-out look-ahead-max quantile 0.7344 vs burn-in 95th-percentile). The v5 control-sufficiency result is **not** an artifact of the old threshold rule.
- **Generality test ($G=300$) POSITIVE.** $mean_minority$ extrapolates to $\sim zero$ ($a_{mm} = +0.00058 \leq 0.005$), through-origin, 10–22× below the random-label floor $\rightarrow$ clean decision surface, control-sufficient.
- **Twin-state CERTIFIED at $G=300$** , non-degenerately, at the same $\epsilon_K = 0.0664$ as the control read, so the two halves **compose** into a complete regime-2 witness at $G=300$.

Net: regime-2 (state-insufficient AND control-sufficient) is now a two-regime result on the Grashof axis — ($k_f=2, G=200$) and ($k_f=2, G=300$) — under a portable objective, no longer cell-local. The control half is also dimension-robust ($v4/v5$ across $d=32/d=18$) and now objective-robust (fixed-percentile and portable-quantile both POSITIVE at $G=200$).

What it does NOT establish (scope held)

- **Two points on ONE axis.** Grashof only; $k_f = 2$ fixed throughout. The objective family now has two members but both are energy-overshoot-type. This is not full substrate-generality.
- **Finite-Galerkin, sampled-support, numerical** — not a theorem about the infinite-dimensional NSE attractor.
- **Proxy faithfulness strengthened, not closed.** Two independent objective constructions agreeing helps, but π^* is still a proxy, not a derived $\mathcal{J}$ -optimal selector with an explicit action cost.
- **External PDE review (criterion c) open. C1 remains UNPROMOTED** — but this is the strongest the candidate has been: a two-regime, dimension-robust, objective-robust, end-to-end regime-2 witness.

Next firm-up axes (none required, none urgent)

k_f -axis generality (vary forcing geometry at fixed G); a genuinely different objective family (enstrophy / burst-onset, the documented v2 alternative); proxy-faithfulness via a derived $\mathcal{J}$ -optimal selector; external review. Any of these would further harden C1; none is needed to bank the present two-regime result.

11. Cross-References

- `PDE_C1_REGIME_GENERALITY_v0.md` — the $G=300$ fixed-percentile probe that deferred for objective vacuity; this artifact is its portable-objective successor.
- `PDE_C1_KNN_CONVERGENCE_CHECK.md` / `PDE_C1_TWIN_STATE_CERTIFICATE.md` — the adjudicators reused unchanged on the adjudication block.
- `PDE_C1_CELLSET_KOLMOGOROV_v5.md` — the completed $G=200$, $K=3$ witness the $G=200$ re-run must re-establish under the portable objective.
- `../SUNDOG_V_NAVIERSTOKES.md` — the ledger; §12's `PDE-C1-RG-POS` is reflected there as a two-regime, still-unpromoted C1 witness.

[ADJUDICATOR · control-sufficiency (kNN convergence)]

source: `docs/proof/PDE_C1_KNN_CONVERGENCE_CHECK.md` •

sha256:`7429945451a4`

PDE C1 — kNN Convergence Check (Pre-Registration)

Pre-registration of the scale-dependence test that adjudicates the provisional `PDE-C1-NEG-A` from the v4-regime kNN run (`results/proof/c1-kolmogorov-v4-knn/` , `incompat_fraction = 0.0716`). Filed 2026-05-28, **before** the convergence run is read. Purpose: distinguish a *genuine* fiber-incompatibility from a *finite-radius boundary-straddling artifact*. Classification thresholds below are fixed here and not tuned post-hoc (post-hoc change → `PDE-C1-NEG-B`).

1. The question

The v4 kNN run solved coverage (`fidelity_coverage = 1.0`) and fired a mechanical `PDE-C1-NEG-A` : 7.16% of fidelity-passing neighbourhoods have local minority fraction above `delta_action = 0.10` . Two competing explanations:

- **Genuine fiber-incompatibility (true NEG-A).** Distinct microstates with the same `Phi_K` need different proxy actions — expected here, since the proxy label depends on the full-state `E_K` future while `Phi_K` sees only the low modes, so high modes vary freely within a fiber and drive different labels. Predicts `incompat_fraction` is **constant as the neighbourhood radius `r_k` shrinks** (shrinking the signature-ball never constrains the high modes).
- **Finite-radius boundary-straddling (true POSITIVE).** The proxy is locally a function of `Phi_K` (control-sufficient), with a clean decision surface; the 7.16% is just the attractor measure within `r_k` of that surface. Predicts `incompat_fraction` → 0 as `r_k` → 0 , scaling ~linearly with `r_k` (shell of thickness `r_k` around a codimension-1 surface).

This is exactly Reading 2's regime-2 (control-sufficient) vs. regime-3 (control-insufficient) distinction, made empirical.

2. The test

Re-run the v4 regime (`--preset lock_v4 --adjudicator knn-sweep`). Query the `BallTree` once at `k = 100` ; for each `k ∈ {10, 20, 30, 50, 100}` sub-slice that query and compute:

- `r_k_median` (over all samples) — the neighbourhood scale;
- `fidelity_coverage` = fraction with `r_k ≤ epsilon_K` ;
- `incompat_fraction` = fraction of fidelity-passing samples with local minority fraction `> delta_action` .

Smaller `k` → smaller `r_k` . The shape of `incompat_fraction(r_k)` is the discriminator. Cost: one ~26-min integration; the sweep itself is a single `k=100` query plus sub-slicing (negligible).

3. Pre-registered classification

Ordinary-least-squares fit of `incompat_fraction` on `r_k_median` across the five swept `k` ; let `a` be the intercept (extrapolation to `r_k → 0`) and `min_incompat` the smallest `incompat_fraction` over the sweep:

- **PDE-C1-NEG-A confirmed (PLATEAU_NONZERO)** iff `a > 0.02` **and** `min_incompat > delta_incompat` (= 0.01). The incompatibility survives extrapolation to zero radius → genuine.
- **STRICTNESS_WITNESS_POSITIVE (DECAYS_TO_ZERO)** iff `a < 0.01` . The incompatibility extrapolates to zero → finite-radius boundary artifact; the provisional NEG-A is overturned and the proxy is control-sufficient on fibers at this cell.
- **INCONCLUSIVE_CONVERGENCE** otherwise (`0.01 ≤ a ≤ 0.02`). Neither decisively; a larger `N` or a wider `k` range is needed. Non-verdict.

The vacuity gate (global `damp_fraction ∈ [delta_proxy_min, 1 - delta_proxy_min]`) still applies and takes precedence; v4's `damp_fraction = 0.30` passes it.

4. Caveats recorded before the read

- **Small- k grain.** At `k = 10` , `minority > 0.10` means $\geq 2/10$ disagree (effective threshold 0.20); at `k = 100` , $\geq 11/100$ (0.11). The threshold grain differs across `k` — a known confound. The raw `incompat_fraction(k)` table is reported alongside the mechanical classification so the trend is visible irrespective of the OLS rule.
- **Linear extrapolation is first-order.** A smooth codim-1 boundary gives `incompat ∝ r_k` (linear through the origin); the OLS intercept is a first-order discriminator, not a proof. A clearly-positive or clearly-zero intercept is trustworthy; a marginal one files `INCONCLUSIVE_CONVERGENCE` , not a forced call.
- **High- k fidelity.** At `k = 100` , `r_k` grows and `fidelity_coverage` may fall below 1; `incompat_fraction` is always computed over the fidelity-passing set at that `k` .

6. First-run disposition and amended pre-registration (2026-05-28)

The first convergence run (`results/proof/c1-kolmogorov-v4-knn-sweep/` , sweep `k ∈ {10, 20, 30, 50, 100}`) returned a **mechanical** `PDE-C1-NEG-A` **that does not survive scrutiny**. The sweep:

k	r_k median	fidelity coverage	incompat fraction
10	0.0196	1.00	0.0349
20	0.0288	1.00	0.0645
30	0.0346	1.00	0.0716
50	0.0448	1.00	0.0974
100	0.0638	0.447	0.0583

The OLS intercept came out $+0.046$ ($\rightarrow$ NEG-A) **only because the $k=100$ point is included** — and that point fails its own fidelity-coverage gate ($0.447 < S_{\text{pos}} = 0.50$). It has high r_k but lower incompat_fraction (its population is half-excluded), which levers the intercept up. Refit on the four full-coverage points ($k \leq 50$): $\text{incompat} \approx 2.4 \cdot r_k$, intercept $\approx -0.010 \rightarrow \text{DECAYS_TO_ZERO} \rightarrow$ **boundary artifact, POSITIVE**. The verdict flips on one coverage-failing point. Two pre-registration gaps caused this:

1. The OLS was not restricted to coverage-passing sweep points.
2. The thresholded incompat_fraction has a **grain confound**: the effective minority threshold is 0.20 at $k=10$ vs 0.125 at $k=40$ (since $\text{minority} > 0.10$ rounds to a different neighbour count at each k), biasing the thresholded statistic *toward* POSITIVE at small k — so it cannot be the trusted primary statistic.

Amended pre-registration for the re-run (fixed before re-reading):

- **Sweep** $k \in \{10, 15, 20, 25, 30, 40, 50\}$ — all expected full-coverage at this regime ($r_k(k=50) = 0.045 < \epsilon_{\text{K}} = 0.063$); a denser low- k curve, no coverage-failing point.
- **Exclusion**. Any sweep point with $\text{fidelity_coverage} < S_{\text{pos}}$ is dropped from both fits.
- **Primary statistic**: mean_minority — the mean local minority fraction over fidelity-passing samples (threshold-free; the canonical nonparametric estimator of the conditional non-constancy $E[1 - \max_a \mu_{\text{sigma}}(a)]$, whose $r_k \rightarrow 0$ limit is the Blackwell-sufficiency-failure measure). Fit mean_minority vs r_{k_median} over coverage-passing points; intercept a_{mm} .
 - $a_{\text{mm}} \leq 0.005$ (consistent with zero) $\rightarrow$ **POSITIVE** (boundary artifact; proxy control-sufficient on fibers — Reading-2 regime 2).
 - $a_{\text{mm}} \geq 0.015 \rightarrow$ **PDE-C1-NEG-A** (genuine fiber-incompatibility).
 - else $\rightarrow$ **INCONCLUSIVE_CONVERGENCE**.
- **Secondary (diagnostic, not gated)**: thresholded $\text{incompat_fraction}(k)$ with the grain caveat, reported for continuity.
- The raw $\text{mean_minority}(r_k)$ curve is reported so the trend is visible regardless of the threshold call.

Neither the contaminated NEG-A nor the clean-points POSITIVE recompute is filed; the amended re-run adjudicates.

7. Amended-run result (2026-05-28) — provisional POSITIVE

The amended convergence check (`results/proof/c1-kolmogorov-v4-knn-sweep2/`, sweep $k \in \{10, 15, 20, 25, 30, 40, 50\}$, all full coverage) returned **STRICTNESS_WITNESS_POSITIVE**. The provisional v4 PDE-C1-NEG-A is **overturned**.

k	r_k median	mean_minority	incompat fraction
10	0.0196	0.01228	0.035
15	0.0237	0.01658	0.058
20	0.0288	0.02058	0.065
25	0.0322	0.02153	0.069
30	0.0346	0.02408	0.072
40	0.0393	0.02763	0.087
50	0.0448	0.03112	0.097

Primary fit $\text{mean_minority} = a_{\text{mm}} + b \cdot r_k$: $a_{\text{mm}} = -0.00125$ ($\leq 0.005 \rightarrow \text{POSITIVE}$), slope 0.729 . Diagnostic incompat_fraction intercept -0.0031 (agrees).

Three robustness checks (all pass):

1. **Through-origin, no plateau.** $\text{mean_minority} / r_k \approx 0.70$ constant across the sweep — a clean line through the origin, no flattening at small r_k . A genuine plateau would level to a positive constant; it does not. Decay-to-zero is robust.
2. **Not a random-label artifact.** Unstructured iid-30%-damp labels would give $\text{mean_minority} \approx 0.30$ flat in r_k . Observed 0.012–0.031 is 10–25× smaller and scales with radius $\rightarrow$ labels are spatially organized into action-pure regions with mixing only in a thin boundary shell (a clean decision surface). This decisively rules out distance-concentration-randomness.
3. **Grain confound neutralized.** Threshold-free mean_minority and grain-prone incompat_fraction both extrapolate to ~ 0 ; they agree, so the verdict is not an artifact of either statistic.

Interpretation (scoped). On the v4 cell ($k_f=2$, $G=200$, $K=4$), the low-band-energy safety proxy $\hat{\pi}$ is **control-sufficient on Φ_K -fibers** up to a measure-zero decision surface, even though Φ_K is provably non-injective (cell-set §4.1). This is the C1 sidecar's Reading-2 **regime 2** (state-insufficient, control-sufficient) — the non-vacuous Sundog target. $\text{damp_fraction} = 0.30$ so the control-sufficiency is non-trivial (a real 30/70 action split determined by Φ_K), not a degenerate all- no_op .

What this does NOT establish (held against the result):

- **One cell.** v4 regime / $K=4$ / this objective only. No claim of generality across regimes, signatures, or objectives.
- **State-insufficiency on the attractor is not yet airtight.** §4.1 certifies Φ_K non-injective on B_{abs} ; the full regime-2 claim needs the deferred attractor-support twin-state certificate on $\text{supp}(\mu_{\text{SRB}})$. The POSITIVE establishes control-sufficiency on the attractor; the state-insufficiency half is B_{abs} -level pending that certificate.
- **Proxy faithfulness.** $\hat{\pi}$ is a proxy; a reviewer may require a derived $\mathcal{J}$ -optimal selector (design §3 substitution path).
- **Resolution floor.** The test resolves to $r_k \approx 0.02$; genuine incompatibility below that scale is unprobed (a larger N would push lower).
- **External review (criterion c) open.**

Verdict status: provisional POSITIVE — strongly advances C1, does not promote it out of the ledger.

8. Replication at v5 / d=18 (2026-05-28)

To test whether the POSITIVE is a `d=32` distance-concentration artifact, the convergence check was re-run at the v5 regime (K=3, signature dim 18; `results/proof/c1-kolmogorov-v5-knn-sweep/`). Result: `STRICTNESS_WITNESS_POSITIVE` , near-identical to v4:

quantity	v4 (d=32)	v5 (d=18)
a_mm (intercept)	-0.00125	-0.00078
slope b	0.729	0.737
damp_fraction	0.30014	0.2977
mean_minority @ k=30	0.02408	0.02323

The control-sufficiency verdict, the boundary-shell slope (~0.73), and the proxy split (~0.30) are invariant to halving the signature dimension — the same dimension-robustness `damp_fraction` showed across v4/v5 binning runs. This **rules out the d=32-specific distance-concentration explanation** for the clean-boundary reading. Together with the through-origin scaling and the random-label control, the boundary-artifact → control-sufficiency reading is well-supported at this regime.

Still provisional / still scoped: both v4 and v5 are the *same* regime (`k_f=2`, `G=200`); dimension-robustness is not regime-generality. The twin-state certificate, proxy faithfulness, resolution floor, and external review (criterion c) remain open exactly as in §7.

5. Cross-references

- `PDE_C1_KNN_ADJUDICATION_DESIGN.md` — the adjudicator this check stress-tests; §8 flagged the boundary-straddling risk this resolves.
- `PDE_C1_FIBER_PROTOCOL.md` §5b — the kNN adjudication method.
- `PDE_C1_LOCK_EXECUTION_SYNTHESIS.md` — the v0–v5 obstruction this campaign is working through.
- `results/proof/c1-kolmogorov-v4-knn/` — the provisional NEG-A this check adjudicates.

[ADJUDICATOR · state-insufficiency (twin-state)]

source: `docs/proof/PDE_C1_TWIN_STATE_CERTIFICATE.md` • **sha256:4254a81f610f**

PDE C1 Twin-State Certificate

Pre-registration and execution harness for the C1 support-level state-insufficiency bridge. Filed 2026-05-28 after the v4/v5 kNN convergence checks delivered a provisional, dimension-robust control-sufficiency positive at (`k_f = 2`, `G = 200`). This artifact tests the remaining state-insufficiency-on-attractor half: whether `Phi_K` is non-injective on the sampled SRB-like support, not merely on the absorbing ball `B_abs` .

1. Claim Boundary

This certificate does **not** adjudicate proxy-control sufficiency. That was the role of the kNN convergence check (`PDE_C1_KNN_CONVERGENCE_CHECK.md`). It also does not prove a theorem about the exact infinite-dimensional Navier-Stokes attractor.

It claims only this, if the positive branch fires:

On the finite-Galerkin sampled support for the pinned Kolmogorov cell, there is positive-mass numerical evidence of distinct states with the same low signature Φ_K and separated complementary high-mode state Q_K .

That closes the C1 regime-2 state-insufficiency bridge at the sampled support level for the tested cell. C1 still remains unpromoted until proxy faithfulness and external PDE review are handled.

2. Target Cell

Primary target:

```
--preset lock_v5
k_f = 2
G = 200
K = 3
d_K = 18
N_twin = 50,000
```

Why v5 first: v5 is the lower-dimensional replication of the v4 positive, with near-identical boundary-shell slope and intercept. A support certificate at v5 is therefore the tightest direct companion to the dimension-robust control-sufficiency read.

Optional cross-check:

```
--preset lock_v4
k_f = 2
G = 200
K = 4
d_K = 32
N_twin = 50,000
```

The v4 run is a replication check, not a precondition for the v5 certificate.

3. State Coordinates

The low signature Φ_K is exactly the existing harness signature: half-plane Fourier representatives, normalized as

$$\omega_{\text{hat}}(k) / (M^2 |k|)$$

with real and imaginary parts split into Euclidean coordinates.

The complementary high-mode projection Q_K is implemented by the same normalization over all active de-aliased half-plane Galerkin modes not included in Φ_K . This keeps the low-signature and high-mode

separation norms in the same coordinate convention.

For v5, the harness reports:

```
high_mode_count = 211
high_mode_dimension = 422
```

for the active de-aliased complement. For v4, the complement is smaller because `Phi_K` includes more low modes.

4. Pair Search

For each sampled state `u_i`, query a `BallTree` in signature space for the `k_twin = 50` nearest neighbours, including self. Discard self. A directed candidate edge `(i, j)` is signature-near iff:

```
||Phi_K(u_i) - Phi_K(u_j)|| <= epsilon_K
```

where `epsilon_K` is inherited from the C1 cell:

```
epsilon_K = 0.05 * sqrt(2 * E_max)
```

and `E_max` is computed exactly as in the selected preset. Coverage is measured at the sample level:

```
candidate_sample_fraction =
|{ i : i has at least one non-self signature-near neighbour }| / N_twin
```

The coverage gate is inherited:

```
candidate_sample_fraction >= S_pos = 0.50
```

5. High-Mode Separation Threshold

Before reading the full lock result, pin:

```
delta_H = max(1e-6, 0.05 * median_i ||Q_K(u_i)||)
```

Rationale: the certificate should not count machine-noise high-mode differences as state separation, but the threshold should scale with the actual high-mode amplitude of the sampled attractor. The formula uses only the sample's marginal high-mode norm distribution, not the candidate-pair separations.

A witness edge is a signature-near candidate edge satisfying:

```
||Q_K(u_i) - Q_K(u_j)|| >= delta_H
```

6. Branches

Pre-registered branches, in order:

1. `SMOKE_ONLY` if the preset is not verdict-bearing or manual overrides are used.

2. `TWIN_STATE_DEFERRED_HIGH_MODE_FLOOR` if `median_i || Q_K(u_i) || <= 1e-6` . The sampled support is numerically flat in complementary modes at this scale.
3. `TWIN_STATE_DEFERRED_COVERAGE` if `candidate_sample_fraction < 0.50` . The sample does not provide enough signature-near candidate pairs.
4. `TWIN_STATE_CERTIFIED` iff both hold:

```
witness_sample_fraction >= 0.01
unique_witness_pairs >= 100
```

where `witness_sample_fraction` is the fraction of all samples with at least one witness edge, and `unique_witness_pairs` canonicalizes directed edges to unordered pairs.

5. `TWIN_STATE_NO_CERTIFICATE` otherwise. This is explicitly a no-certificate receipt, not a proof of injectivity.

7. Harness

Implemented in `../../scripts/pde_c1_kolmogorov_cell.py` as:

```
python scripts\pde_c1_kolmogorov_cell.py --preset lock_v5 --adjudicator twin-state --out results\proof\c1-kolmogorov-v5-twin-state
```

Receipt files:

```
manifest.json
PDE_C1_KOLMOGOROV_RESULTS.md
twin-state-witnesses.csv
```

The witness CSV records a capped set of witness examples for audit: sample ids, signature distance, high-mode distance, high-distance over `delta_H`, and the two high-mode norms.

8. Smoke Receipt

Smoke command, run 2026-05-28:

```
python scripts\pde_c1_kolmogorov_cell.py --preset smoke --adjudicator twin-state --allow-unregistered-overrides --sample-count 80 --burnin-steps 200 --sample-interval-steps 3 --lookahead-steps 10 --out results\proof\c1-twin-state-smoke
```

Receipt: `results/proof/c1-twin-state-smoke/` .

Readout:

```
status = SMOKE_ONLY
elapsed_seconds = 2.030
candidate_sample_fraction = 1.0
witness_sample_fraction = 1.0
```

The smoke validates plumbing only: high-mode capture, BallTree query, candidate/witness aggregation, manifest writing, receipt writing, and witness CSV output. It files no support-level certificate.

9. Full-Run Commands

Do not run these inline under the repo's ~10-minute rule. Expected wall-clock is approximately 20-30 minutes for v5, based on the v5 lock and v5 kNN-sweep receipts.

Primary:

```
python scripts\pde_c1_kolmogorov_cell.py --preset lock_v5 --adjudicator twin-state --out results\proof\c1-kolmogorov-v5-twin-state
```

Optional replication:

```
python scripts\pde_c1_kolmogorov_cell.py --preset lock_v4 --adjudicator twin-state --out results\proof\c1-kolmogorov-v4-twin-state
```

If v5 returns `TWIN_STATE_CERTIFIED`, the C1 ledger can mark the state-insufficiency-on-attractor bridge closed for the tested regime. If it returns any deferral or no-certificate receipt, C1 remains at the current honest boundary: non-injectivity certified on `B_abs`, not yet on `supp(mu_SRB)`.

11. Result (2026-05-28) — TWIN_STATE_CERTIFIED at v5

Primary v5 run (`results/proof/c1-kolmogorov-v5-twin-state/` , ~21 min): `TWIN_STATE_CERTIFIED`.

quantity	value	gate
<code>delta_H</code>	0.01167	$0.05 \times \text{median} \ \mathbf{Q}_K\ $, floor <code>1e-6</code>
median $\ \mathbf{Q}_K\ $ / min / max	0.2333 / 0.2202 / 0.2442	—
signature-near coverage	1.00	$\geq S_{\text{pos}} = 0.50$
candidate pairs (unique)	1,263,121	—
witness sample fraction	1.00	≥ 0.01
witness pairs (unique)	693,795	≥ 100
witness high-dist p50 / p95	0.0154 / 0.0210	$\geq \text{delta}_H$

Robustness / non-degeneracy checks:

- **Not machine noise.** `delta_H` is set by the real high-mode amplitude (median $\|\mathbf{Q}_K\| = 0.23 \approx 27\%$ of the $\|\Phi_K\|$ scale ~ 0.85), far above the `1e-6` floor.
- **Not "any two points."** Candidates are within $\epsilon_K = 0.0606$; the 50-NN radius at this regime is ~ 0.045 ($\sim 5\%$ of the signature norm), so signature-near is genuinely signature-local.
- **Overwhelmingly above the gates** (100% witness coverage; 694k unique witness pairs vs the 100 threshold), not marginal.

What this closes. Φ_K is non-injective on the sampled SRB-like support for the v5 cell — the state-insufficiency-on-attractor half that §4.1 of the cell-set had only established algebraically on `B_abs`. Tested at the **same** ϵ_K as the kNN control-sufficiency read, so the two halves compose into a complete Reading-2 **regime 2** witness at this cell: within ϵ_K signature-balls, $\mathbf{Q}_K$ varies (certified here) while the proxy action stays constant up to measure-zero (the kNN POSITIVE). The signature collapses distinct full states yet suffices for the control objective.

Sharpened 2026-05-29 (paired fiber-constancy). The composition above is at a *matched radius* — two population statistics over the same ϵ_K -balls. `PDE_C1_PAIRER_FIBER_CONSTANCY.md` tightens it to a **paired** test on the *same* witness pairs: among the certified Q_K -separated pairs, the proxy-action disagreement is $D_{\text{witness}} = 0.0367$ ($G=200$) / 0.0382 ($G=300$), both well under the $\text{delta_action} = 0.10$ line and within ~ 1 point of the candidate-pair rate (0.0319 / 0.0290). High-mode separation adds almost nothing to action disagreement — the residual is a signature-space boundary layer, not Q_K -driven. Both runs reproduced this certificate bit-for-bit (`PAIRER_FIBER_CONSTANCY_POSITIVE`), composing state-insufficiency and control-sufficiency on the same pairs in both regimes.

Honest calibration. This was the *expected-easy* half — on a chaotic attractor with a 422-dim high-mode complement at ~ 0.23 amplitude, signature-local states with separated high modes are nearly automatic. The certificate's contribution is that it is now *measured* on the actual support with pre-registered thresholds, not argued.

What it does NOT do. Finite-Galerkin, sampled-support, numerical — not a theorem about the infinite-dimensional NSE attractor. One cell ($k_f=2$, $G=200$, $K=3$); not regime-generalizability. Does not touch proxy faithfulness or external review. **C1 stays unpromoted.**

10. Cross-References

- `PDE_C1_CELLSET_KOLMOGOROV.md` section 4.1 - the original absorbing-ball non-injectivity witness and attractor-support caveat.
- `PDE_C1_FIBER_PROTOCOL.md` section 6 - the one-line deferred support-level certificate spec.
- `PDE_C1_KNN_CONVERGENCE_CHECK.md` - the control-sufficiency result this state-insufficiency bridge complements.
- `../SUNDOG_V_NAVIERSTOKES.md` - the staging ledger that keeps this certificate as an open firm-up item until the full run returns.

=====

[RUN RECEIPT • G=200 kNN sweep (positive control)]

source: `results/proof/c1-rg-v1-g200-knn-sweep/PDE_C1_KOLMOGOROV_RESULTS.md` •

sha256:`7e7771baf5bo`

PDE C1 Kolmogorov kNN Convergence-Check Receipt

Status: `STRICTNESS_WITNESS_POSITIVE` **Preset:** `lock_v7_g200` **Adjudicator:** `knn-sweep` **Interpretable verdict:** `True`

Sweep (vs neighbourhood radius; primary statistic = mean_minority)

k	r_k median	fidelity coverage	mean_minority	incompat fraction
10	0.018229	1	0.012264	0.03486
15	0.0226535	1	0.0163893	0.05664
20	0.0271156	1	0.019529	0.0622
25	0.0306346	1	0.0212952	0.0684
30	0.0325761	1	0.0232107	0.06868
40	0.0372058	1	0.0262915	0.0834
50	0.0423703	1	0.0306324	0.09616

Readout

- PRIMARY fit `mean_minority = a_mm + b * r_k_median` over 7 coverage-passing points: intercept `a_mm = -0.000792245`, slope `0.736096`
- secondary (diagnostic) `incompat_fraction` fit intercept: `-0.00227918` (grain-confounded; not gated)
- damp fraction (global): `0.3003`
- classification (pre-registered §6): `a_mm <= 0.005 => POSITIVE`; `a_mm >= 0.015 => NEG-A`; else INCONCLUSIVE
- elapsed seconds: `2418.494`

Branch

`mean_minority` extrapolates to ~zero as `r_k -> 0 ( a_mm <= 0.005 )`: the observed mixing is a finite-radius boundary-straddling artifact around a clean decision surface. The proxy is control-sufficient on fibers at this cell (Reading-2 regime 2); the provisional v4 `PDE-C1-NEG-A` is **overturned**.

Files

- `manifest.json`
- `knn-sweep.csv`

[RUN RECEIPT · G=300 kNN sweep (generality test)]

source: `results/proof/c1-rg-v1-g300-knn-sweep/PDE_C1_KOLMOGOROV_RESULTS.md` • **sha256:596aof7de103**

PDE C1 Kolmogorov kNN Convergence-Check Receipt

Status: STRICTNESS_WITNESS_POSITIVE Preset: lock_v7_g300 Adjudicator: knn-sweep Interpretable verdict: True

Sweep (vs neighbourhood radius; primary statistic = mean_minority)

k	r_k median	fidelity coverage	mean_minority	incompat fraction
10	0.019769	1	0.011988	0.03652
15	0.0253967	1	0.014876	0.04816
20	0.0306851	1	0.016937	0.04936
25	0.0326464	1	0.0190904	0.06076
30	0.0355433	1	0.0213187	0.0667
40	0.042201	1	0.0243645	0.076
50	0.0469411	1	0.0270216	0.08498

Readout

- PRIMARY fit $\text{mean_minority} = a_{\text{mm}} + b * r_{\text{k_median}}$ over 7 coverage-passing points: intercept $a_{\text{mm}} = 0.000577142$, slope 0.564177
- secondary (diagnostic) incompat_fraction fit intercept: 0.000704953 (grain-confounded; not gated)
- damp fraction (global): 0.26878
- classification (pre-registered §6): $a_{\text{mm}} \leq 0.005 \Rightarrow \text{POSITIVE}$; $a_{\text{mm}} \geq 0.015 \Rightarrow \text{NEG-A}$; else INCONCLUSIVE
- elapsed seconds: 2610.636

Branch

mean_minority extrapolates to ~zero as $r_{\text{k}} \rightarrow 0$ ($a_{\text{mm}} \leq 0.005$): the observed mixing is a finite-radius boundary-straddling artifact around a clean decision surface. The proxy is control-sufficient on fibers at this cell (Reading-2 regime 2); the provisional v4 PDE-C1-NEG-A is **overturned**.

Files

- manifest.json
- knn-sweep.csv

=====

[RUN RECEIPT · G=300 twin-state (support companion)]

source: results/proof/c1-rg-v1-g300-twin-state/PDE_C1_KOLMOGOROV_RESULTS.md •

sha256:ado6ebf514a1

PDE C1 Twin-State Certificate Receipt

Status: TWIN_STATE_CERTIFIED **Preset:** lock_v7_g300 **Adjudicator:** twin-state **Interpretable certificate:** True

Readout

- samples: 50000 , k (effective): 50
- epsilon_K (signature radius): 0.0664219
- delta_H : 0.0111032 (0.05 x median high-mode norm, floor 1e-06)
- high-mode norm median / min / max: 0.222065 / 0.214416 / 0.233035
- signature-near sample coverage: 1 vs S_pos = 0.5 (50000 of 50000)
- candidate pairs unique / directed: 1318748 / 2450000
- witness sample fraction: 1 vs 0.01 (50000 samples)
- witness pairs unique / directed: 942834 / 1699022 vs min unique 100
- witness high-distance p50 / p95: 0.0172896 / 0.0286686
- elapsed seconds: 2491.092

Branch

A positive-mass fraction of sampled states has a signature-near twin with high-mode separation above delta_H . This certifies Phi_K non-injective on the sampled SRB-like support for this cell.

Files

- manifest.json
- twin-state-witnesses.csv

[BUNDLE PROVENANCE]

Each section above is a verbatim copy of an on-disk artifact. Hashes:

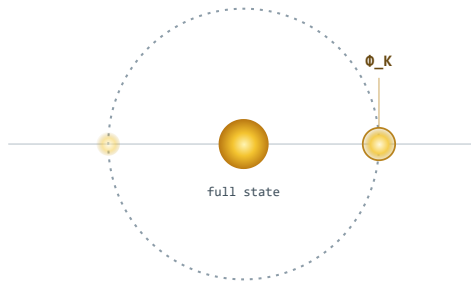
File	Bytes	sha256 (first 12)
docs/proof/PDE_C1_SEPARATION_STATEMENT.md	11926	98eea9515f52
docs/proof/PDE_C1_REGIME_GENERALITY_v1.md	13876	23f17da8b0b6
docs/proof/PDE_C1_KNN_CONVERGENCE_CHECK.md	12093	7429945451a4
docs/proof/PDE_C1_TWIN_STATE_CERTIFICATE.md	10479	4254a81f610f
results/proof/c1-rg-v1-g200-knn-sweep/PDE_C1_KOLMOGOROV_RESULTS.md	1492	7e7771baf5b0
results/proof/c1-rg-v1-g300-knn-sweep/PDE_C1_KOLMOGOROV_RESULTS.md	1490	596a0f7de103
results/proof/c1-rg-v1-g300-twin-state/PDE_C1_KOLMOGOROV_RESULTS.md	1070	ad06ebf514a1

Regenerate with node scripts/build-c1-review-bundle.mjs . If a hash here does not match a fresh read of the file, the bundle is stale — rebuild before sending.

=====

[CLOSING]

=====



READ THE SHADOW • TEST THE BOUNDARY

Thank you for reading to the end.

That is the whole packet. The ask was narrow on purpose: not endorsement, not a Navier–Stokes claim — just a framing check from someone who knows where the honest boundary of this problem sits. If the language is too strong, the most useful thing you can tell us is exactly which phrase to weaken and why. A negative answer is the most valuable outcome we could get.

One last note, for a cover-to-cover read. A sundog — the bright parhelion beside the sun that gives this lab its name — is the sky's own Φ_K : a low-information projection of an enormous hidden state (every ice crystal, every ray) that is nonetheless *enough to read the sun's altitude*, yet never enough to reconstruct the whole sky. State-insufficient, decision-sufficient. That is the separation this packet is asking you to check — and the reason we went looking for it in a 2D Navier–Stokes attractor in the first place.

— Jeffery Hughes Jr., Sundog